# EMV®

# 3-D Secure Approval

# Test Harness for Split-SDK as System Under Test

Version 1.15

11 Jul 2025

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# Revision Log

| Version | Date | Description |
|---|---|---|
| 1.0 | 05 May 2022 | Initial version: mainly extracts from former [TEST_REQ_SDK] v2.5 and updates to manage Shell-SDK and Browser-SDK Test Architecture:<br>• Section "3. Communication Test Requirements for Split-SDK/Native and Shell as SUT" extracted from [TEST_REQ_SDK] v2.5<br>  ○ Only editorial changes<br>  ○ Section is also applicable for Shell-SDK<br>• New section "4. Communication Test Requirements for Split-SDK/Browser as SUT"<br>• "5. API Specifications " extracted from [TEST_REQ_SDK] v2.5<br>  ○ Editorial changes<br>  ○ API description have been made more generic to manage also Shell-SDK and Browser-SDK Test Architecture. |
| 1.1 | 10 Jun 2022 | P3DS_TP-374: Clarifications / Typo corrections:<br>• "5.2.3 – Tool.3DSSplitSdk_API.fetchTC"<br>  ○ Data elements: clarification, it may be defined upon TPP or SUT needs<br>  ○ Output parameters: clarification, it contains the visualEvidencePoint<br>  ○ RunTcId Presence: Not present in HTTP request, but returned in HTTP Response<br>• "5.2.4.1 – pErq elements, Visual Evidence Endpoint" : clarifications on the manaqement of the reception of pErq |
| 1.2 | 22 Jul 2022 | Upgrade references to documents due to new 2.3.1 specifications |

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| Version | Date | Description |
|---|---|---|
| 1.3 | 23 Sept 2022 | Upgrade version to referenced documents<br>• [3DS_Core_2.3.1] / [3DS_SDK_Spec_2.3.1] / [3DS_Split_SDK_Spec_2.3.1]<br>• [TEST_REQ_ALL_SUT]<br>• [TEST_REQ_SDK]<br>• New reference to [RFC_4122]<br>Editorial changes to realign Split-SDK Variant terminology from 2.3.1 specifications. |
| 1.4 | 28 Oct 2022 | No content change. Only reformatting to latest EMVCo Template document |
| 1.5 | 25 Nov 2022 | Updated version of referred documents :<br>• [TEST_REQ_ALL_SUT]<br>• [TEST_REQ_SDK] |
| 1.6 | 14 Dec 2022 | Updated version of referred documents :<br>• [TEST_REQ_ALL_SUT]<br>• [TEST_REQ_SDK] |
| 1.7 | 27 Jan 2023 | P3DS_TP-471 - Clarification of the fetchTC usage has been done in section 5.2.3 of the 'Test Harness for Split-SDK' specification.<br><br>P3DS_TP-478 - Sections 3.1.4 and 4.1.2 have been corrected with the correct label i.e. TPP (3DS Server). In addition, the security related to the CReq/CRes functions has been specified in the new sections 3.1.5 and 4.1.3.<br><br>P3DS_TP-480 - Clarification on visualEvidenceEndpoint and new visualEvidenceEndpointURL in section 5.2.3.<br><br>P3DS_TP-484 : Reference of RFC 2388 has been replaced by the 7578<br><br>P3DS_TP-476 : Different header names used: RunTcId vs TCRunID<br><br>Updated version of referred documents :<br>• [TEST_REQ_SDK] |
| 1.8 | 28 April 2023 | Updated version of referred documents :<br>• [TEST_REQ_SDK] |

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| Version | Date | Description |
|---|---|---|
| 1.9 | 30 June 2023 | Updated version of referred documents :<br>• [TEST_REQ_ALL_SUT]<br>• [TEST_REQ_SDK] |
| 1.10 | 21 July 2023 | Updated version of referred documents :<br>• [TEST_REQ_ALL_SUT]<br>• [TEST_REQ_SDK] |
| 1.11 | 27 Oct 2023 | Updated version of referred documents :<br>• [TEST_REQ_ALL_SUT]<br>• [TEST_REQ_SDK] |
| 1.12 | 26 Apr 2024 | Updated version of referred documents :<br>• [TEST_REQ_ALL_SUT]<br>• [TEST_REQ_SDK] |
| 1.13 | 11 Oct 2024 | Update from EMV® Specification Bulletin No. 255 v4 July 2024 : 3DS 2.1.0 is sunsetted<br>• All references to 3DS 2.1.0 are removed<br>Updated version of referred documents :<br>• [TEST_REQ_ALL_SUT]<br>• [TEST_REQ_SDK] |
| 1.14 | 30 Apr 2025 | Updated version of referred document:<br>• [TEST_REQ_SDK] |
| 1.15 | 11 Jul 2025 | Updated version of referred document:<br>• [TEST_REQ_SDK] |

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# 1. References

| Short name | Reference Document | Version / Date |
|---|---|---|
| [TEST_REQ_ALL_SUT] | Test Requirements for all Systems Under Test | v2.14 / 11 Oct 2024 |
| [TEST_REQ_SDK] | Test Requirements for SUT Default SDK and Split SDK | V2.21 / 11 Jul 2025 |
| [3DS_Core_2.2.0] | EMV® 3-D Secure Protocol and Core Functions Specification EMVCo | v2.2.0 / nov-2018 |
| [3DS_SDK_Spec_2.2.0] | EMV® 3-D Secure – SDK Specification EMVCo | v2.2.0 / nov-2018 |
| [3DS_Core_2.3.1] | EMV® 3-D Secure Protocol and Core Functions Specification EMVCo | v2.3.1.1 / May-2023 |
| [3DS_SDK_Spec_2.3.1] | EMV® 3-D Secure – SDK Specification EMVCo | v2.3.1.1 / May-2023 |
| [3DS_Split_SDK_Spec_2.3.1] | EMV® 3-D Secure – Split-SDK Specification EMVCo | v2.3.1.1 / May-2023 |
| [RFC_4122] | A Universally Unique IDentifier (UUID) URN Namespace. https://tools.ietf.org/html/rfc4122 | - |

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# 2. Introduction

This document describes communication test requirements between Test Environment and 3DS Split-SDK as System Under Test (SUT).

It relies on the [TEST_REQ_SDK] document which describes the high-level requirements applicable for the Split-SDK testing.

Communication test requirements may vary depending on the Split-SDK variants:

- Split-SDK/Native and Split-SDK/Shell requirements are described in section 3
- Split-SDK/Browser requirements are described in section 4

A test API has been designed especially for executing the Split-SDK test cases (see section 5).

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# 3. Communication Test Requirements for Split-SDK/Native and Shell as SUT

## 3.1. Security & network connection

### 3.1.1. Standards

To secure the exchanges between all components, the document specifies some standards for the transport layer to use:

| Transport layer | Description |
|---|---|
| **Protocol** | HTTPS |
| **Confidentiality** | TLS ≥ 1.2 |
| **Integrity** | TLS ≥ 1.2 |
| **Authentication** | Server Authenticate / Mutual Authenticate |
| **Authority** | Private CA / Commercial CA |
| **X509** | rfc5280 PKI Certificate and Certificate Revocation List (CRL) Profile |
| **TLS 1.3** | rfc8446 TLS Protocol 1.3 |
| **Certificate Pinning** | If requested by the system under test |

The Test Environment can support the certificate Pining if requested by the SUT for the authentication.

### 3.1.2. User Authentication

The different connections can use specific and additional user authentications. It is optional and depends on the system under test request. This user authentication should support a standard authentication method to assume the TPP has the capability to use. As the interface uses HTTP, the authentication methods should be based on this protocol.

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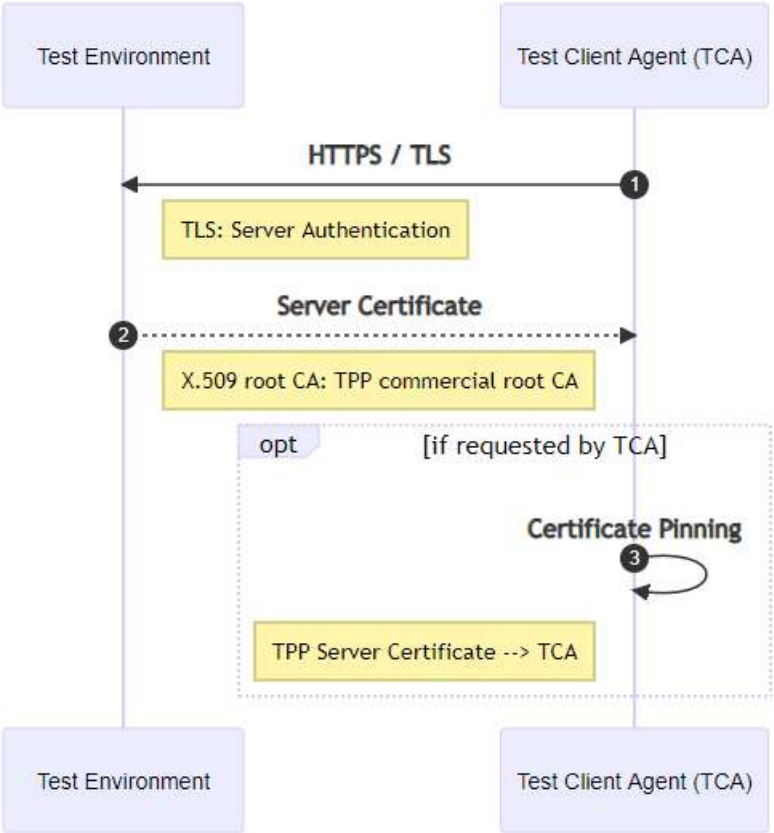
### 3.1.3. Connection from the Test Client Agent to the Test Environment




NOTE: This connection is mainly used for exchanging the fetchTC, pErq and pSrq messages.

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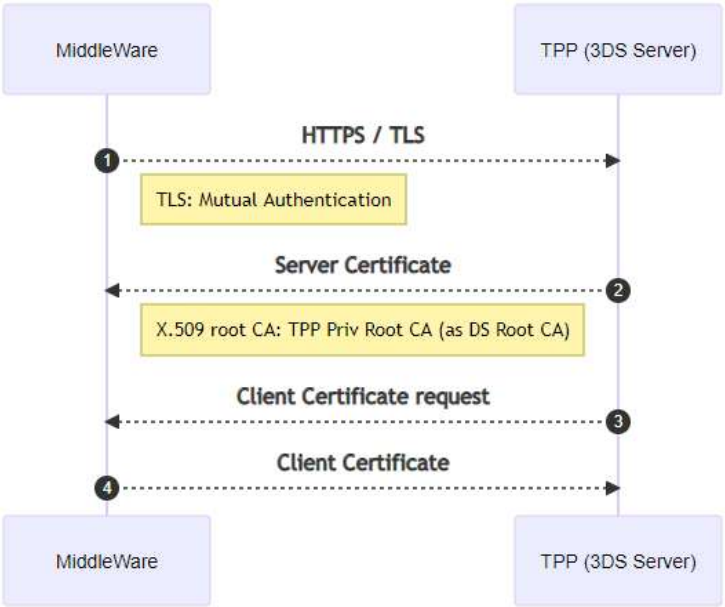
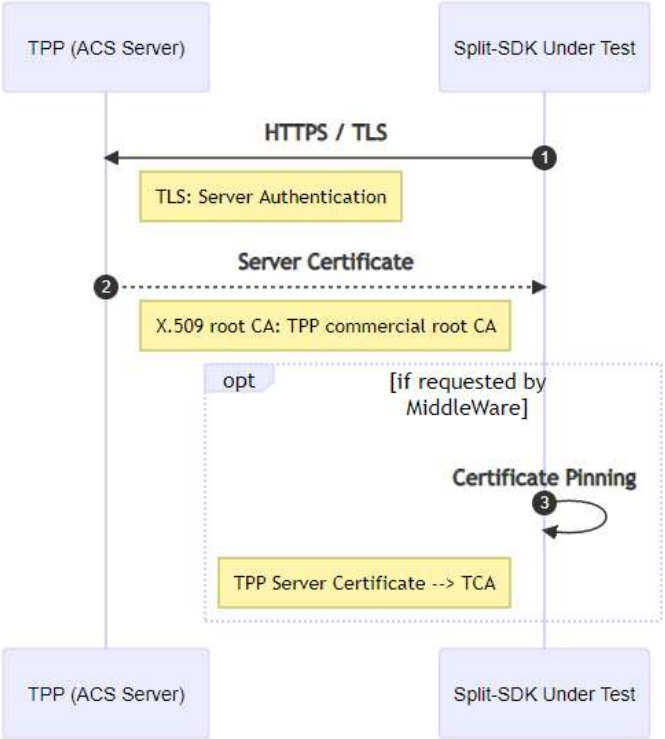
### 3.1.4. Connection from the MiddleWare to the TPP (3DS Server)




NOTE: This connection is mainly used for exchanging the pArq/pArs messages.

### 3.1.5. Connection from the Split-SDK to the TPP (ACS Server)




NOTE: This connection is mainly used for exchanging the cReq/cRes messages.

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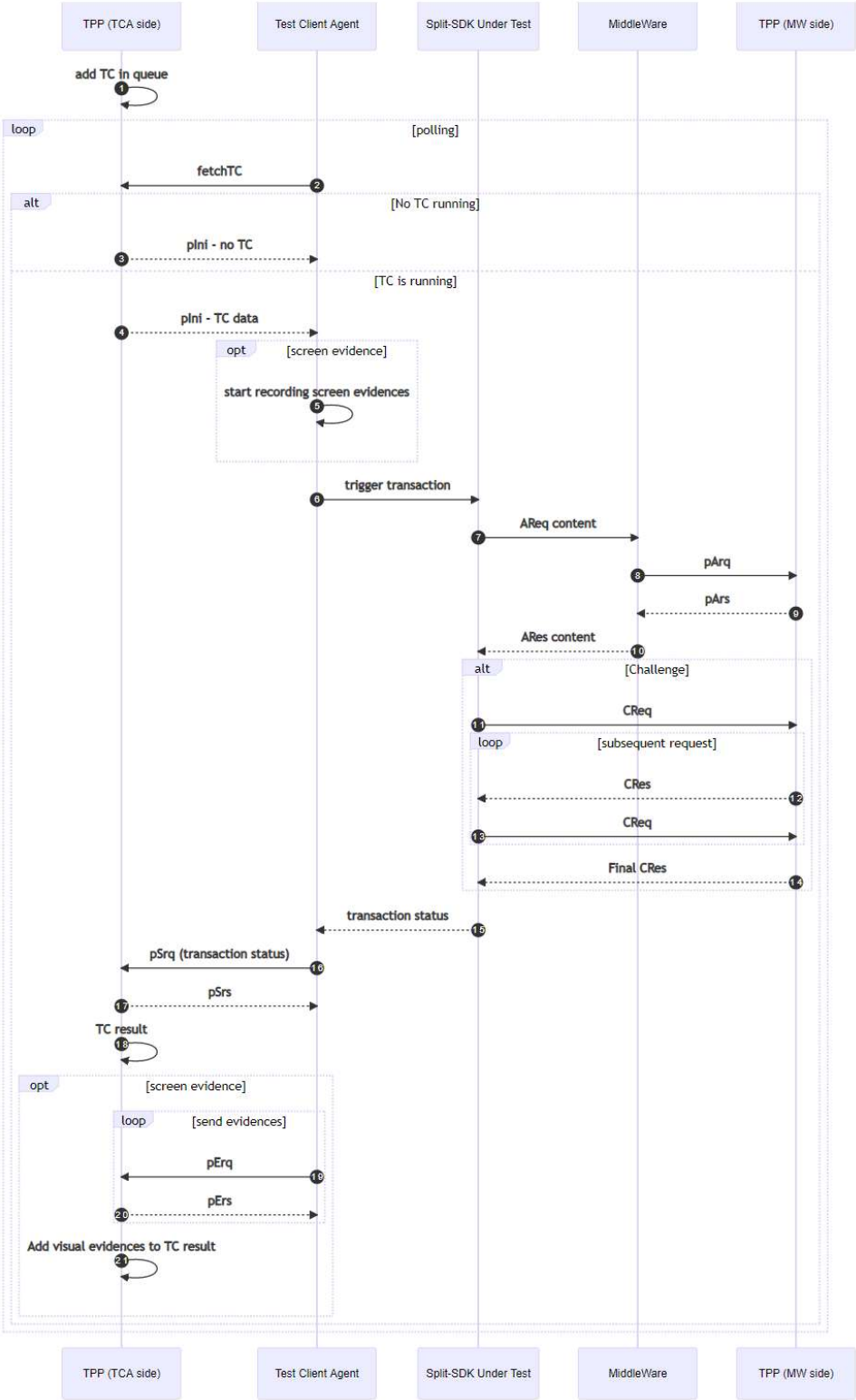
## 3.2. Use cases

### 3.2.1. Nominal flow

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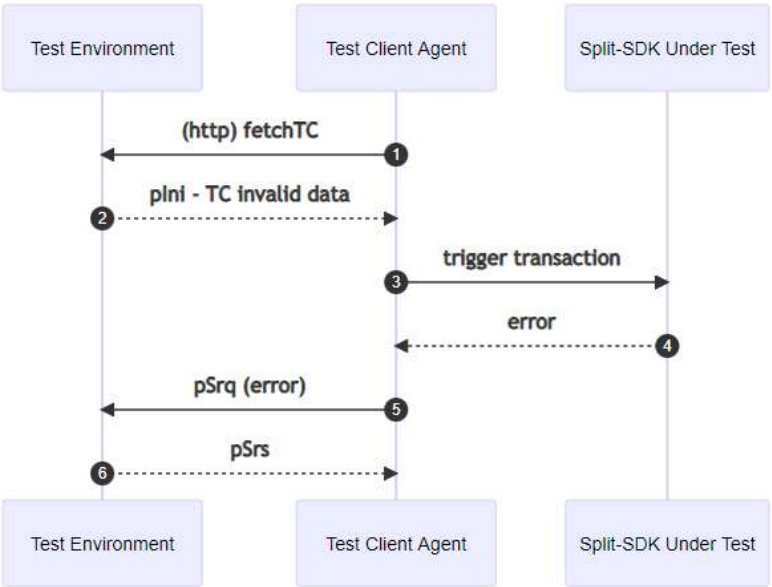
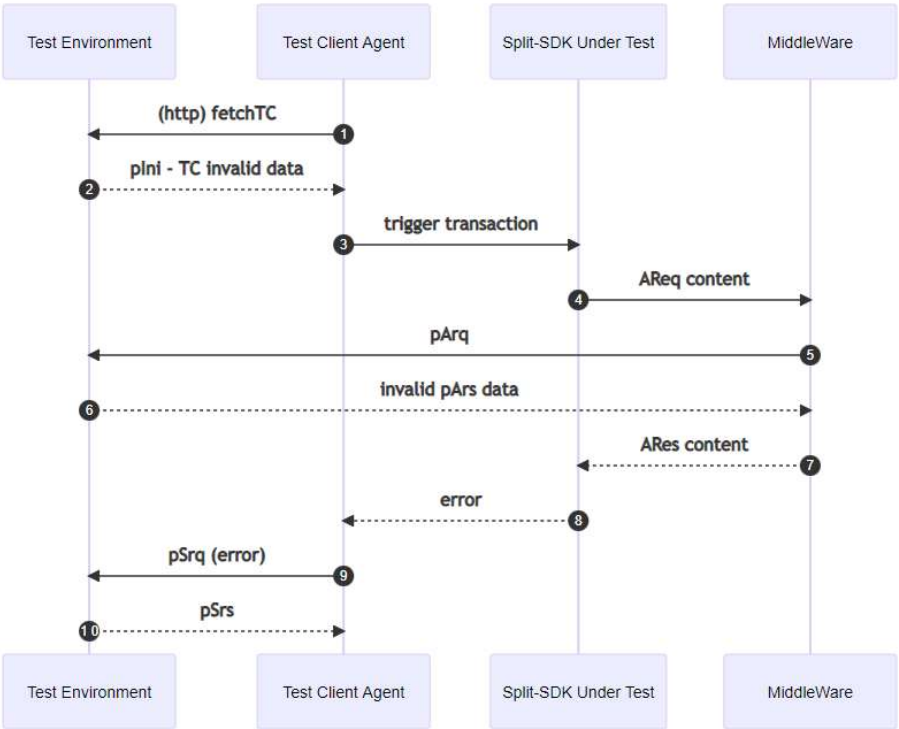
### 3.2.2. Error flow management

To indicate to the TPP that an issue is coming, whatever the issue, some error flows have been created.

#### 3.2.2.1. Trigger transaction error from Test Client Agent to Split-SDK




#### 3.2.2.1. Trigger transaction error from the MiddleWare to the Split-SDK

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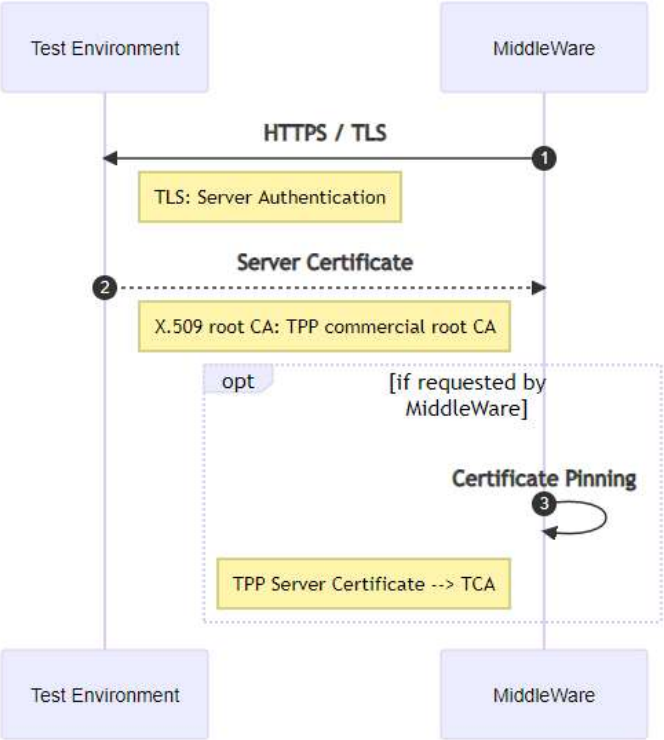
# 4. Communication Test Requirements for Split-SDK/Browser as SUT

## 4.1. Security & network connection

Security standards and user authentications defined in sections 3.1.1 and 3.1.2 apply also for the Split-SDK/Browser.

### 4.1.1. Connection from the MiddleWare to the Test Environment




NOTE: This connection is mainly used for exchanging the fetchTC, pErq and pSrq messages.

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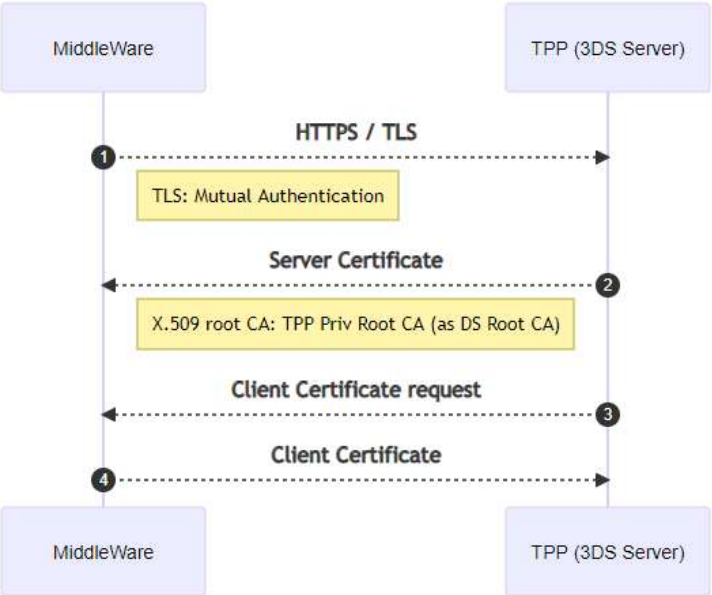
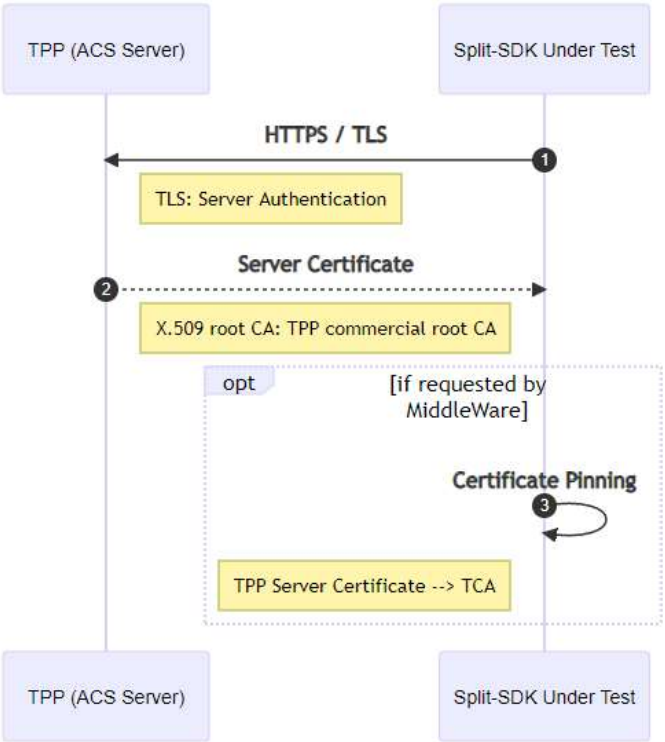
### 4.1.2. Connection from the MiddleWare to the TPP (3DS Server)




NOTE: This connection is mainly used for exchanging the pArq/pArs messages.

### 4.1.3. Connection from the Split-SDK to the TPP (ACS Server)




NOTE: This connection is mainly used for exchanging the cReq/cRes messages.

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### 4.1.4. Connection between the Merchant Web Page and the MiddleWare

The connection between the Merchant Web Page and the MiddleWare remains Product Provider application dependent.

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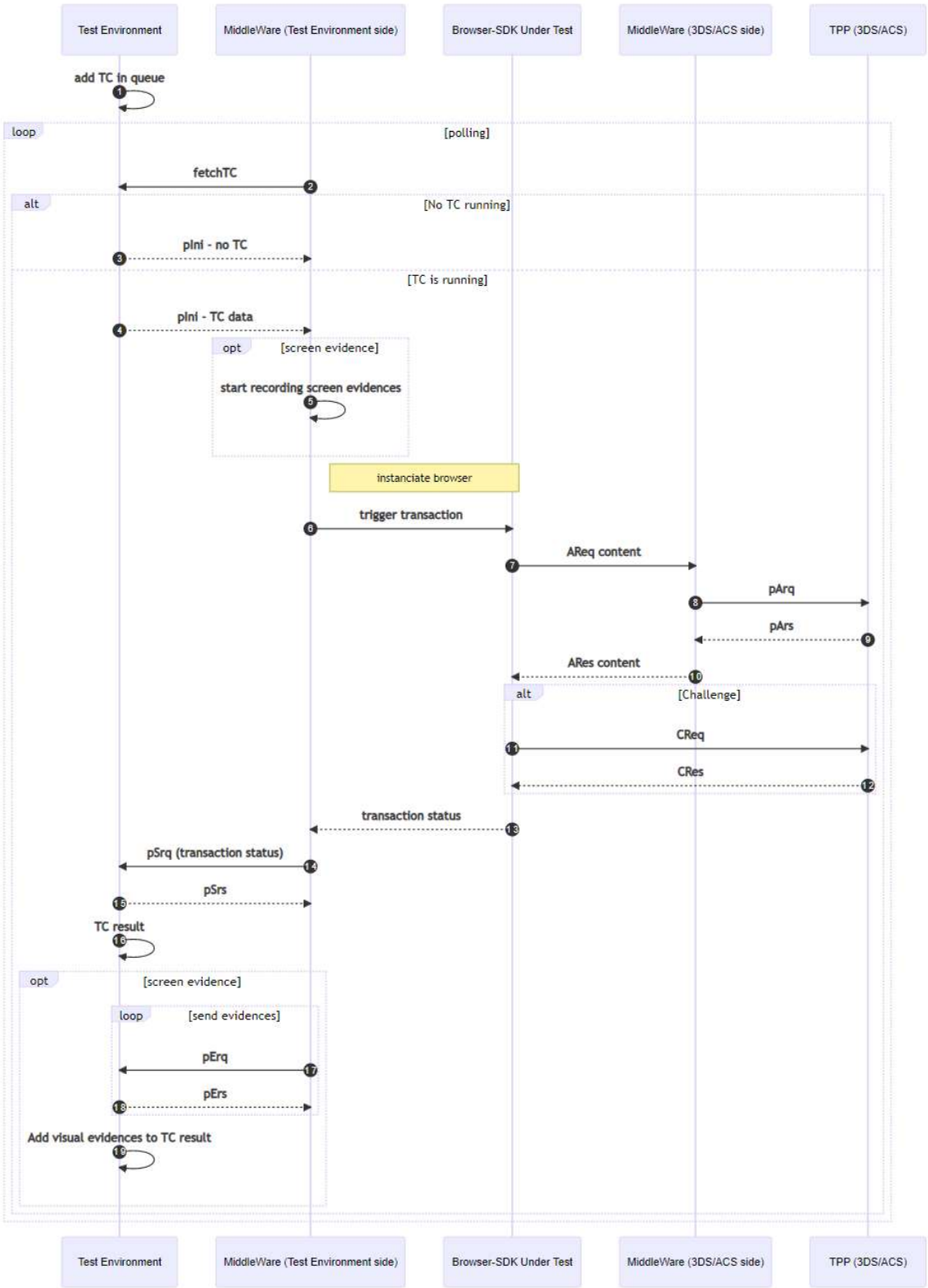
# 4.2. Use cases

## 4.2.1. Nominal flow

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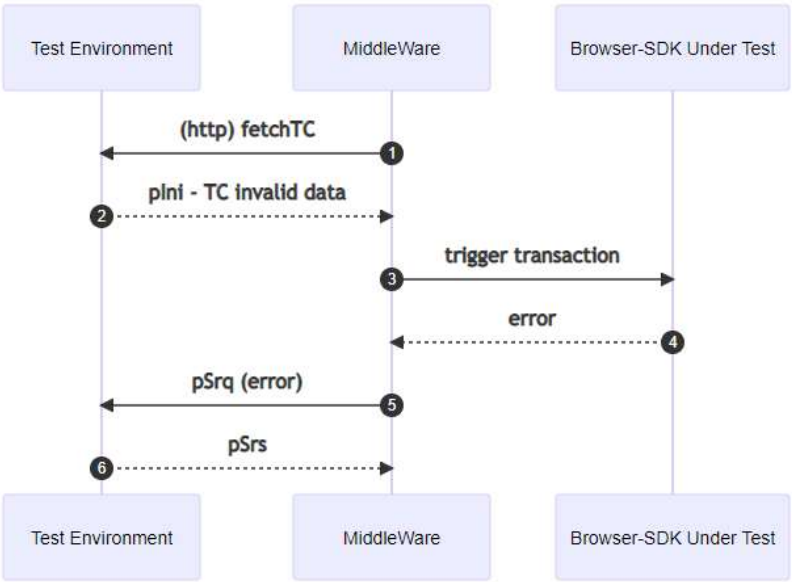
### 4.2.2. Error flow management

To indicate to the TPP that an issue is coming, whatever the issue, some error flows have been created.

#### 4.2.2.1. Trigger transaction error from MiddleWare to Split-SDK/Browser

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# 5. API Specifications

This section describes the specific API messages required for testing the Split-SDK whatever its variant.

| ***Service Name*** | **API exchange** |
|---|---|
| ***Protocol type*** | HTTP (REST) |
| ***Protocol reference*** | rfc7231 Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content |
| ***Protocol Format*** | JSON |
| ***Protocol Format reference*** | rfc7159 JSON Data Interchange Format |
| ***Service Identifier*** | Tool.3DSSplitSdk_API |

## 5.1. RunTcId specific data element

Some specific data elements can be requested by the TPP to execute a test session.

TPP has to retrieve the test case running during a session for a specific System Under Test.

To manage the same process for all SUT, this document specifies this identifier.

The identifier is unique and dynamically generated by the TPP. This ID shall not be related to a specific test case identifier. And the SUT cannot anticipate the test case running based on this identifier. Only the TPP can synchronize this element.

This dynamic run test case identifier shall not be predictive. This ID is named RunTcId.

RunTcId is encoded in UUID4 format as specified in [RFC_4122]. This is a contextual element which shall be exchanged between TPP and SUT.

As this is not a test plan element, this ID should not be present in the test plan messages. This ID shall be present in the HTTP Headers.

The RunTcId shall be present in the following messages

- pIni message
- pArq message
- pArs message
- pSrq message
- pErq message

For the Split-SDK/Native and Split-SDK/Shell, the Test Client Agent has to share the RunTcId with the MiddleWare in order to be able for it to provide it in pArq (the way to share this information between Test Client Agent and Middleware implementation dependent).

The Test environment makes the link between RunTcId and threeDSServerTransID while sending the pArs message.

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Regarding the CReq/CRes messages exchanged between Split-SDK Server and Test Environment, no RunTcId is transmitted neither in message, nor in HTTP header.

From the CReq/CRes messages, the Test environment is able to synchronize them with the current test execution through the field 'threeDSServerTransID'

At the end of the transaction, for the generation of pSrq and pErq messages, the Test Client Agent or the MiddleWare will do the reconciliation with RunTcId (it is implementation dependent)

The Version 4 of UUID is a universally unique identifier that is generated using random numbers.

The Version 4 UUIDs produced by the TPP shall be generated using a secure random number generator.

TPP and SUT can manage this element as a non-standard HTTP Header field.

| *Name* | Description | Example | Status |
|---|---|---|---|
| ***RunTcId*** | Dynamic and Unique identifier<br><br>UUID4 format as specified in [RFC_4122] | `RunTcId: c2cd982a-8a6e-472a-b77f-6a8c089170b3` | Permanent |

## 5.2. Methods

The content of messageType is described previously on this document, the API uses the same data structure as the test plan requires.

### 5.2.1. Tool.3DSSplitSdk_API.pArq

| ***Method Name*** | **pArq** |
|---|---|
| ***Description*** | **proprietary message** |
| ***Caller*** | **Middle Ware** |
| ***Protocol type*** | **HTTP (REST)** |
| ***Protocol method*** | **POST** |
| ***messageType*** | **pArq** |
| ***Data elements*** | **Section 3.2.1 a) of [TEST_REQ_SDK]** |
| ***Output parameters*** | **pArs messageType**<br>**Section 3.2.1 b) of [TEST_REQ_SDK]** |
| ***RunTcId Presence*** | **yes** |

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### 5.2.2. Tool.3DSSplitSdk_API.pSrq

| *Method Name* | pSrq |
|---|---|
| ***Description*** | **proprietary message** |
| ***Caller*** | **Test Client Agent or MiddleWare** |
| ***Protocol type*** | **HTTP (REST)** |
| ***Protocol method*** | **POST** |
| ***messageType*** | **pSrq** |
| ***Data elements*** | **Section 3.1.3 a) of [TEST_REQ_SDK]** |
| ***Output parameters (pSrs)*** | **None (Acknowledge, no data)** |
| ***RunTcId Presence*** | **yes** |

### 5.2.3. Tool.3DSSplitSdk_API.fetchTC

| *Method Name* | fetchTC |
|---|---|
| ***Description*** | **polling request** |
| ***Caller*** | **Test Client Agent or MiddleWare** |
| ***Protocol type*** | **HTTP (REST)** |
| ***Protocol method*** | **GET** |
| ***messageType*** | **NA** |
| ***Data elements*** | **None by default**<br>*(any specific input parameter may be added in accordance with the SUT and TPP providers if required)* |
| ***Output parameters*** | **pIni messageType**<br>**Section 3.1.1 of [TEST_REQ_SDK]**<br>*Plus the 'visualEvidenceEndpoint' indicator if relevant*<br>Or **None (Acknowledge, no data)** if there is no test to be executed |
| ***RunTcId Presence*** | **no**<br>*(not present in the HTTP request but returned in the HTTP Response)* |

The polling requests the TPP to send the pIni element of the current test case.

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The content of the output parameters of the fetchTC method may include some specific data elements for the TPP.

The pIni element is described on the section 3.1.1 of the [TEST_REQ_SDK]. The message includes the challengeUserData as a table of Tool.3DSSplitSdk_API.

In addition to those elements, the messageType pIni should include the elements below in case the SUT supports visual evidence extraction:

- visualEvidenceEndpoint (pErq management)
  - Indicator for TPP to manage endpoint for the visual evidence. These evidences are managed with pErq messageType.

In case there is no test to be executed, the TPP shall send back an empty body response with an HTTP 204 or 200 status code. In such case, the SUT may decide to stop polling the TPP.

It is assumed that, in case a test case consists in executing several transactions, the SUT shall send as many fetchTC requests as transactions to be executed (i.e. one pIni per transaction). In such case, the TPP must use the same RunTcId in each pIni.

When the visualEvidenceEndpoint indicator is set to 'True', the TPP may also provide its root URL implementing the pErq requests, so that the SUT may use it dynamically during the visual evidences sharing:

- visualEvidenceURL (pErq management)
  - Root URL of the TPP server (named as [root_url_endpoint_path] in the [TEST_REQ_SDK]) managing the visual evidences storage.

### 5.2.4. Tool.3DSSplitSdk_API.pErq

| ***Method Name*** | **pErq** |
|---|---|
| ***Description*** | **proprietary message** |
| ***Caller*** | **Test Client Agent or MiddleWare** |
| ***Protocol type*** | **HTTP (REST)** |
| ***Protocol method*** | **Post** |
| ***messageType*** | **pErq** |
| ***Data elements*** | **Visual Evidence Endpoint RFC 7578** |
| ***Output parameters (pErs)*** | **None (Acknowledge, no data)** |
| ***RunTcId Presence*** | **yes** |

#### 5.2.4.1. pErq elements, Visual Evidence Endpoint

If the pIni related to a specific test contains the visualEvidenceEndpoint indicator set to 'true', the SUT has to provide the endpoint element with a pErq messageType.

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The 'visualEvidenceEndpoint' is retrieved after a fetchTC request and the pErq content type is defined in rfc 7578 (Returning Values from Forms: multipart/form-data) as defined in section 3.1.4 of [TEST_REQ_SDK].

An extended time can be allowed to the SUT to send the pErq to the TPP after the pSrq (e.g. in minutes) in order to let the SUT uploading all visual evidences.

Nevertheless, even if the pErq of a related test has not been received yet, it should not prevent the TPP to start a new test sequence. In addition, the TPP shall not wait the pErq message in case the 'visualEvidenceEndpoint' indicator was set to 'false'.

## 5.3. Error Cases

| *http code* | **Message** |
|---|---|
| ***400*** | Service not recognized |
| ***401*** | Client not authenticated |
| ***403*** | Client authenticated but not allowed to use this service |
| ***404*** | Client authenticated but application not known |
| ***405*** | Client authenticated but method not allowed for this application |
| ***408*** | The request is no longer expected - delay (e.g. 15 min) reached since the pSrq |

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